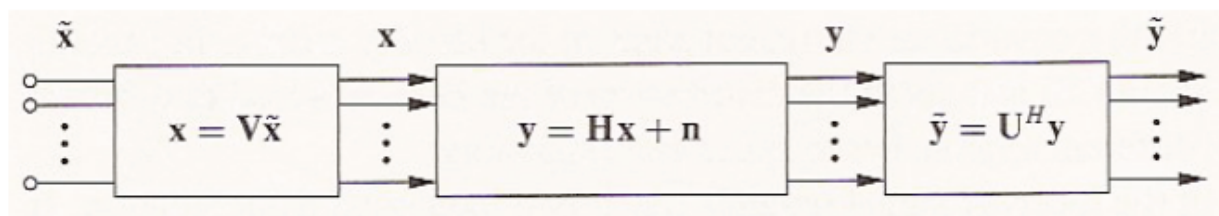


Wireless Digital Communications

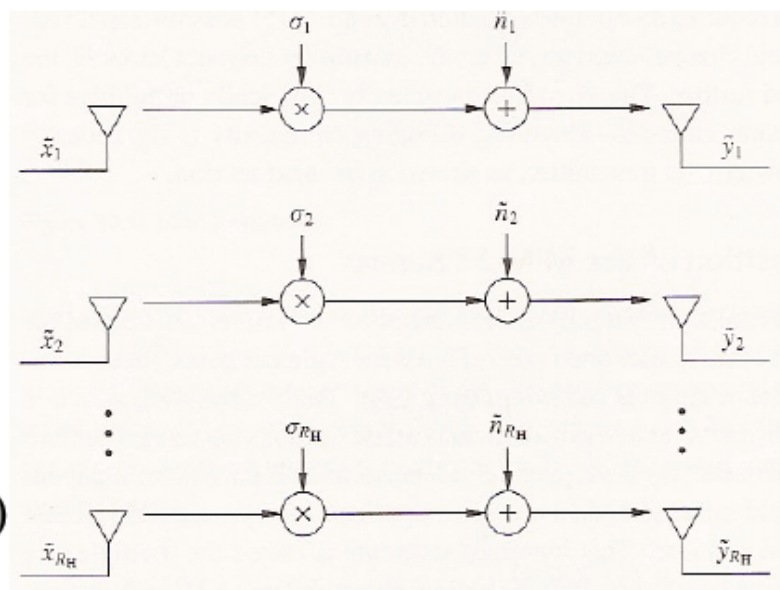
Topic 7 MIMO Transceiver

Transceiver with CSIT and CSIR



Data rate of MIMO
 R_H times data rate
 of SISO with the same
 Bandwidth.

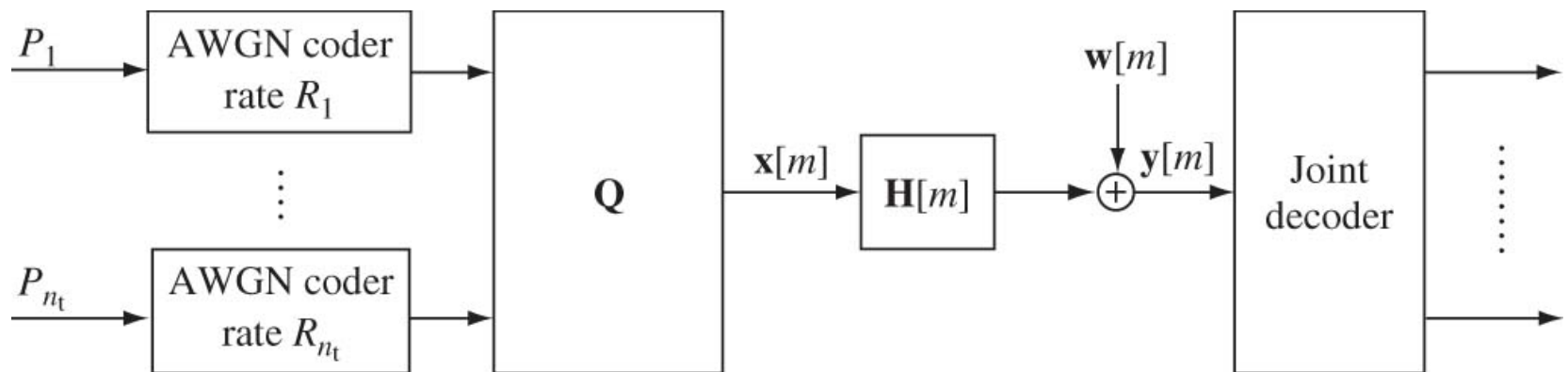
full rank: $R_H = \min(M_r, M_t)$
 linear increase in capacity



$$\tilde{y} = \sigma \tilde{x} + \tilde{n} \quad \hat{\tilde{x}} = \arg \min \left| \tilde{y} - \sigma \tilde{x} \right|$$

Transceiver with CSIR only

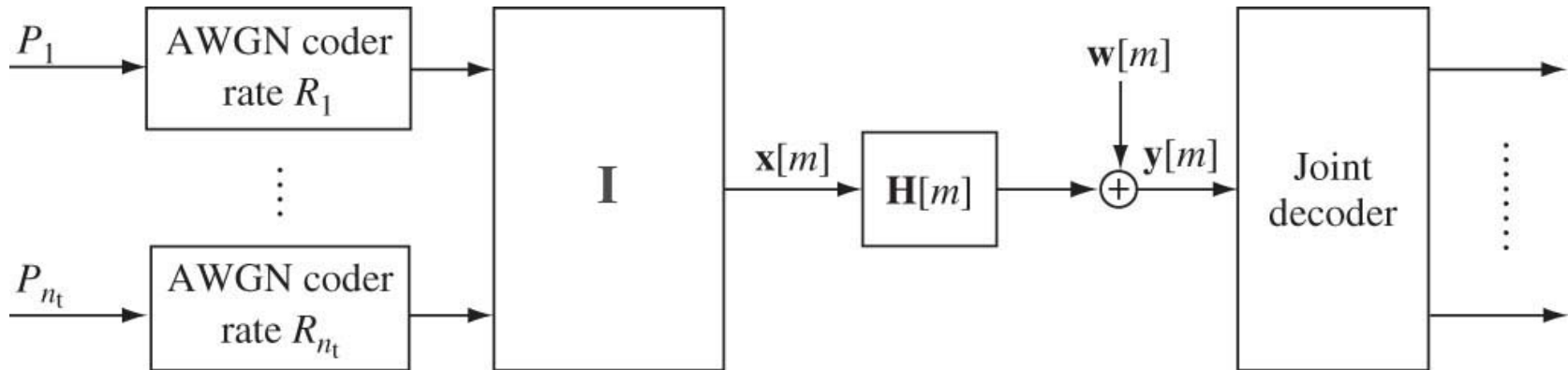
The channel matrix $\mathbf{H}$ and its singular values λ_i^2 's are random and unknown to the transmitter.



Has to fix a $\mathbf{Q}$ and a power allocation independent of $\mathbf{H}$.

$\mathbf{Q}=\mathbf{I}$ and uniform power allocation is optimal in many cases.

Transceiver with CSIR only : V-BLAST



Sending independent coded streams at each of the Tx antennas and joint ML decoding which is too complex.

Optimal Receiver

Idea

Maximum likelihood criterion: return $\hat{\mathbf{x}} = \operatorname{argmax}_{\mathbf{x}} \underbrace{f(\overbrace{\mathbf{H}, \mathbf{y}}^{\text{known at receiver}}, \mathbf{x})}_{\text{likelihood function}}$

Some more details about ML decoder

- Minimum probability of error if $\mathbf{x}$ are equiprobable
- For iid Gaussian noise statistics and equiprobable BPSK signalling, $f(\mathbf{H}, \mathbf{y}, \mathbf{x}) = e^{-c\|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2}$, so

$$\hat{\mathbf{x}} = \operatorname{argmin}_{\mathbf{x} \in \{-1, +1\}^{N_t}} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2$$

Problem

Exponential complexity in N_t

Linear Receiver

- We are interested in the class of linear receiver which computes a decision statistic $\hat{x}_1$ for the first data stream by correlating the received signal $\mathbf{y}$ with an appropriately chosen vector $\mathbf{a}_1$:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{w} = \mathbf{h}_1 x_1 + \sum_{i \neq 1} \mathbf{h}_i x_i + \mathbf{w}$$

$$\hat{x}_1 = \mathbf{a}_1^H \mathbf{y} = (\mathbf{a}_1^H \mathbf{h}_1) x_1 + \sum_{i \neq 1} (\mathbf{a}_1^H \mathbf{h}_i) x_i + \mathbf{a}_1^H \mathbf{w}$$

Linear Receiver :Interference Nulling (Zero-Forcing, Decorrelator)

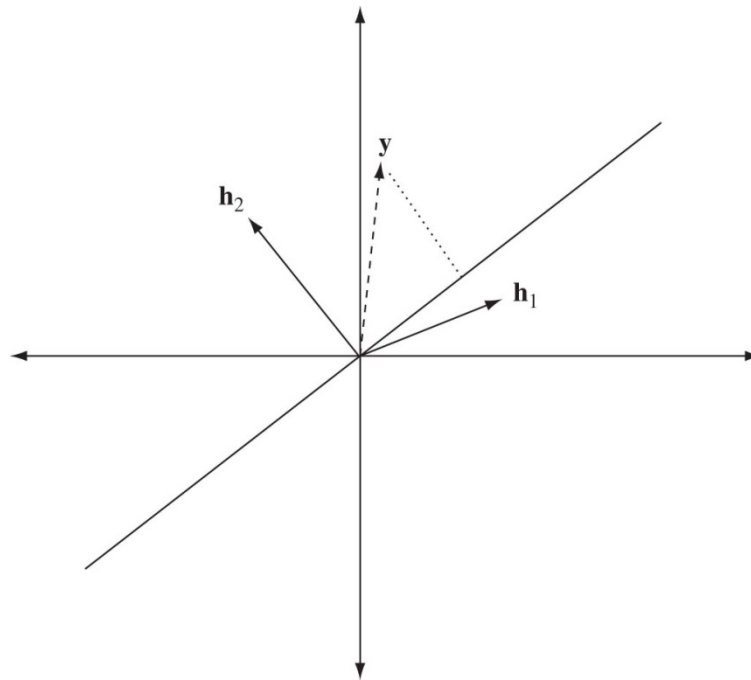
$$\begin{aligned}\mathbf{a}_1^H \mathbf{h}_1 &= 1 \\ \mathbf{a}_1^H \mathbf{h}_2 &= 0 \\ &\vdots \\ \mathbf{a}_1^H \mathbf{h}_{n_t} &= 0\end{aligned}$$

Number of Equations= N_t $\leq$ Number of Variables= N_r

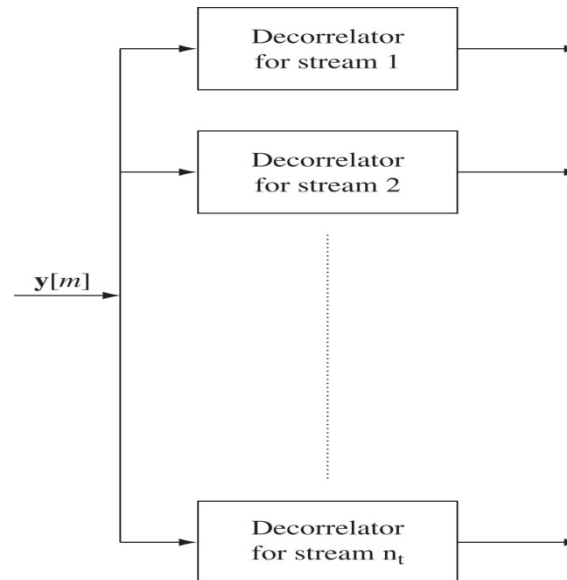
for the existence of solutions

Linear Receiver :Interference Nulling (Zero-Forcing, Decorrelator)

Simple strategy: null out the interference from other antennas.



Receiver Architecture : Bank of Decorrelators

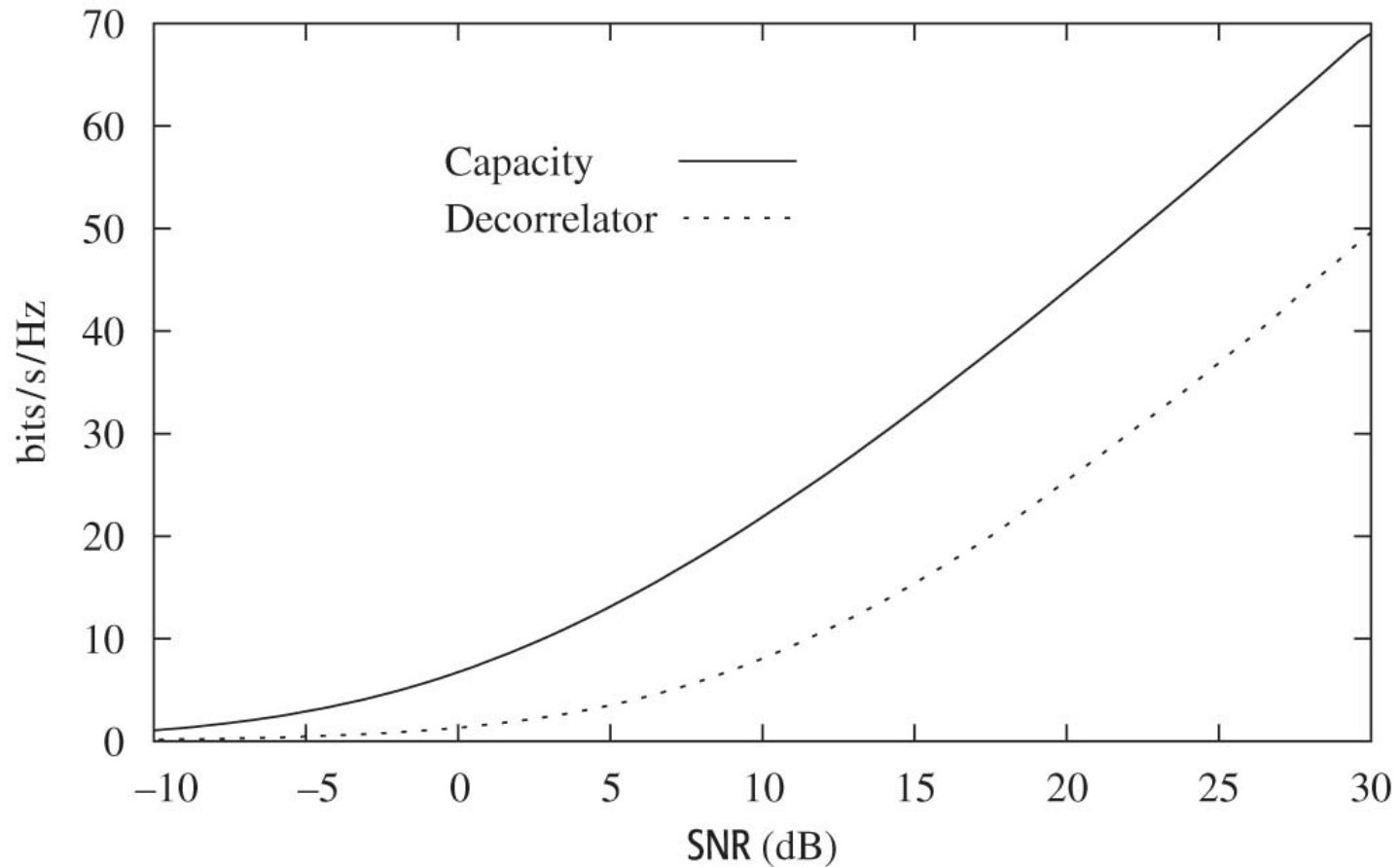


Math

$$\hat{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y}$$

$$\hat{\mathbf{x}} = \mathbf{H}^\dagger \mathbf{y} \text{ where } \mathbf{H}^\dagger = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \text{ if } \mathbf{H} \text{ is "tall"}$$

Bank of Decorrelators: Performance



$n_t = n_r = 8$, i.i.d. Rayleigh

Matched Filter (MF) Receiver: Maximum Ratio Combining (MRC)

$$y = \mathbf{h}_1 x_1 + \sum_{i \neq 1} \mathbf{h}_i x_i + \mathbf{w}$$

$$SNR = \frac{E[|\mathbf{a}_1^H \mathbf{h}_1 x_1|^2]}{E[|\mathbf{a}_1^H \mathbf{w}|^2]} = \frac{|\mathbf{a}_1^H \mathbf{h}_1|^2}{\|\mathbf{a}_1\|^2 \sigma^2} \leq \frac{\|\mathbf{h}_1\|^2}{\sigma^2}$$

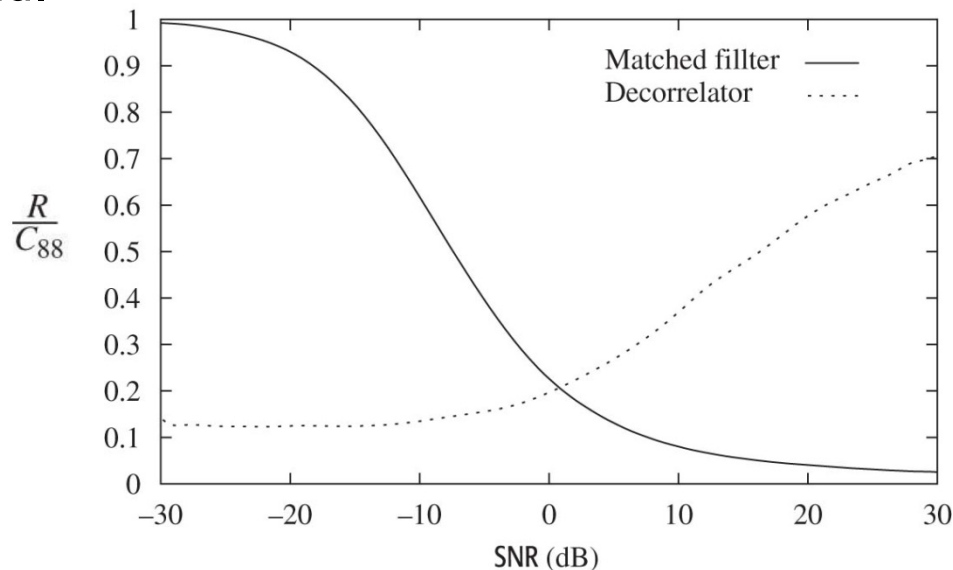
When $\mathbf{a}_1 = \mathbf{h}_1$, SNR is maximized.

Interference Nulling vs Match Filtering

$$y = \mathbf{h}_1 x_1 + \sum_{i \neq 1} \mathbf{h}_i x_i + \mathbf{w}$$

Interference nulling: remove all interference at the expense of reducing the SNR.

Match filtering: projecting onto $\mathbf{h}_1$ to maximize the SNR but SINR may be bad.



Optimal Linear Filter: MMSE

$$y = h_1 x_1 + \sum_{i \neq 1} h_i x_i + w$$

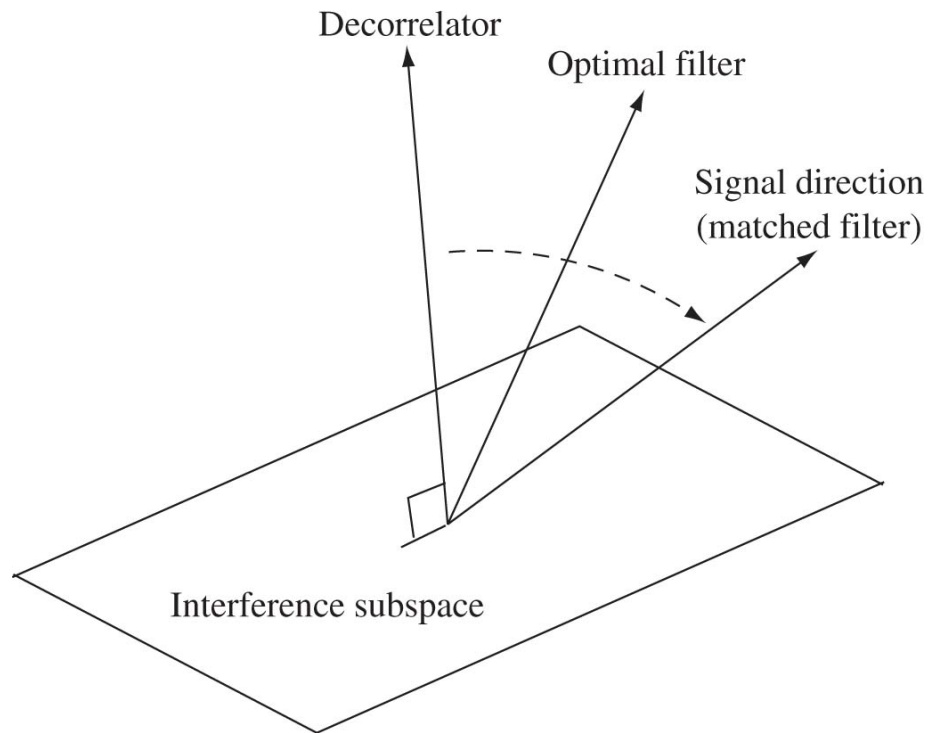
Seek a linear filter that **maximizes the output SINR** at all SNR.

Offers the optimal **compromise** between nulling (Zero-Forcing) and match filtering (Maximum Ratio Combining).

MMSE Receiver

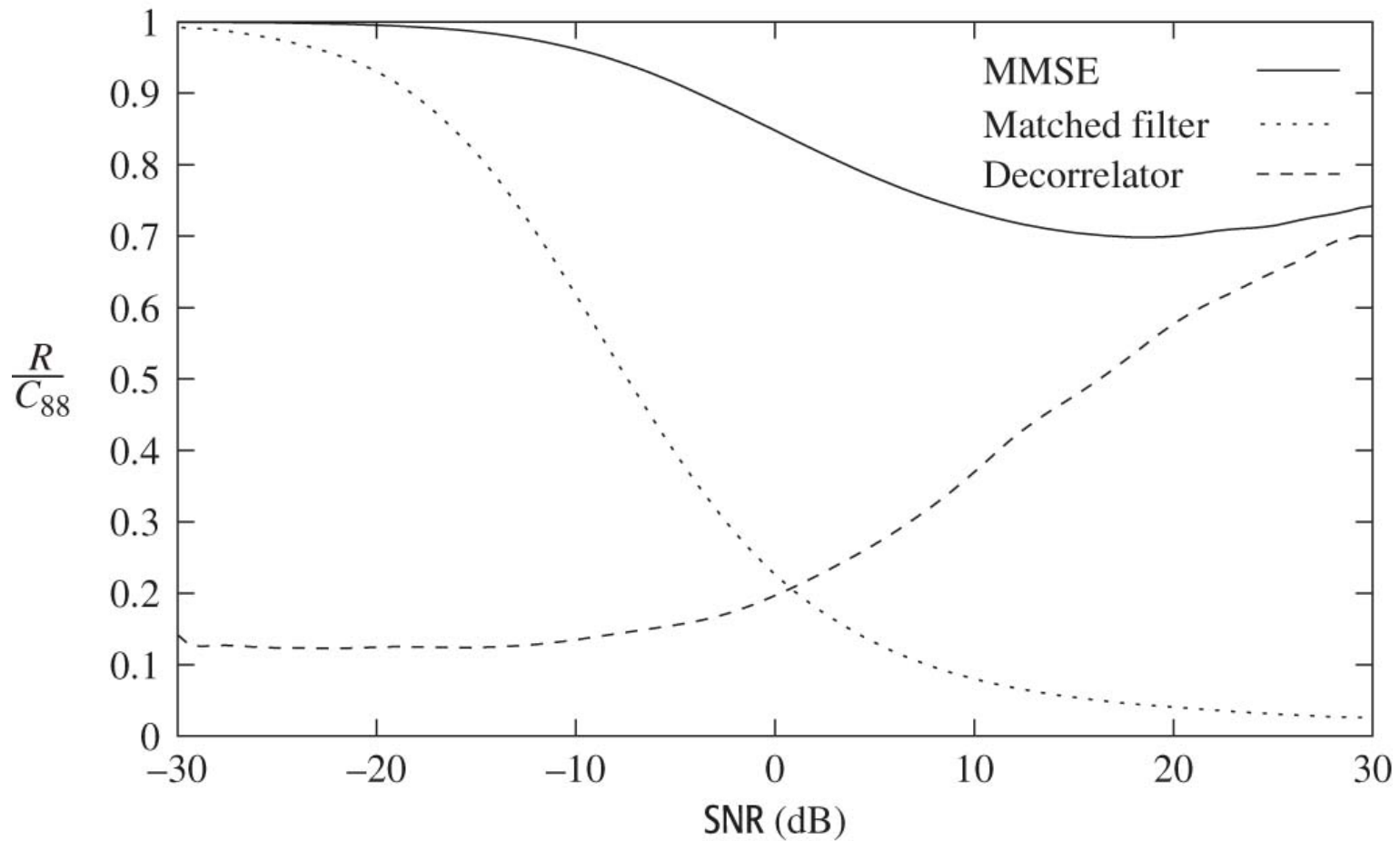
$$\begin{aligned}\mathbf{a}_1 &= \arg \min_{\mathbf{a}} E[|\mathbf{a}^H \mathbf{y} - x_1|^2] \\ &= \arg \min_{\mathbf{a}} \mathbf{a}^H (\mathbf{H}\mathbf{H}^H + \sigma^2 \mathbf{I}) \mathbf{a} - 2\mathbf{a}^H \mathbf{h}_1 + 1 \\ &= (\mathbf{H}\mathbf{H}^H + \sigma^2 \mathbf{I})^{-1} \mathbf{h}_1\end{aligned}$$

MMSE Filter



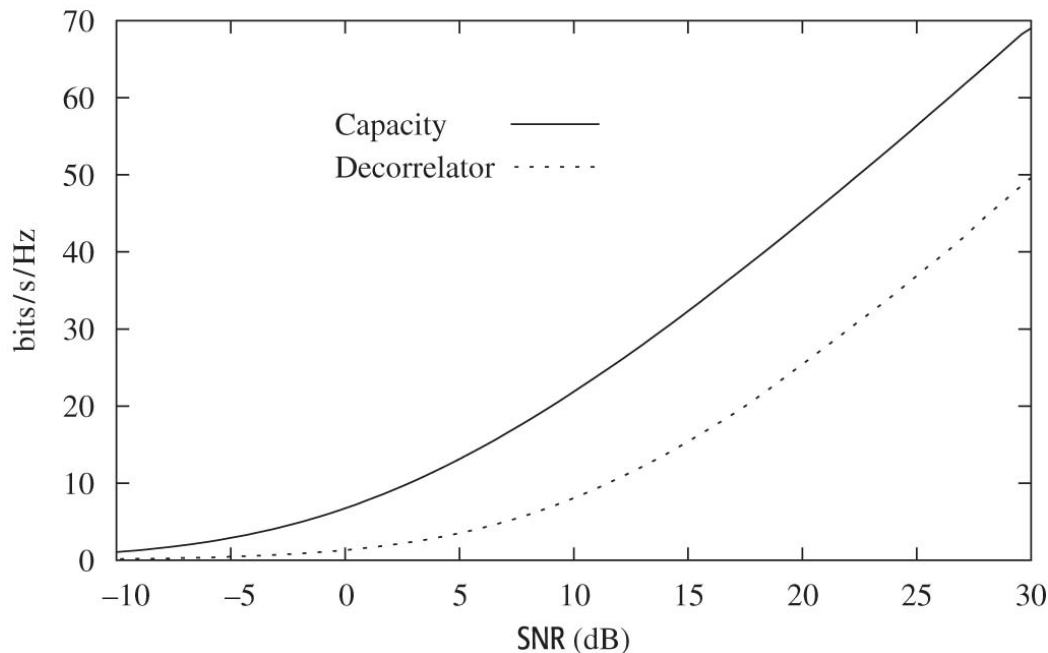
High SNR: MMSE $\rightarrow$ decorrelator
Low SNR: MMSE $\rightarrow$ matched filter

Linear MMSE: Performance

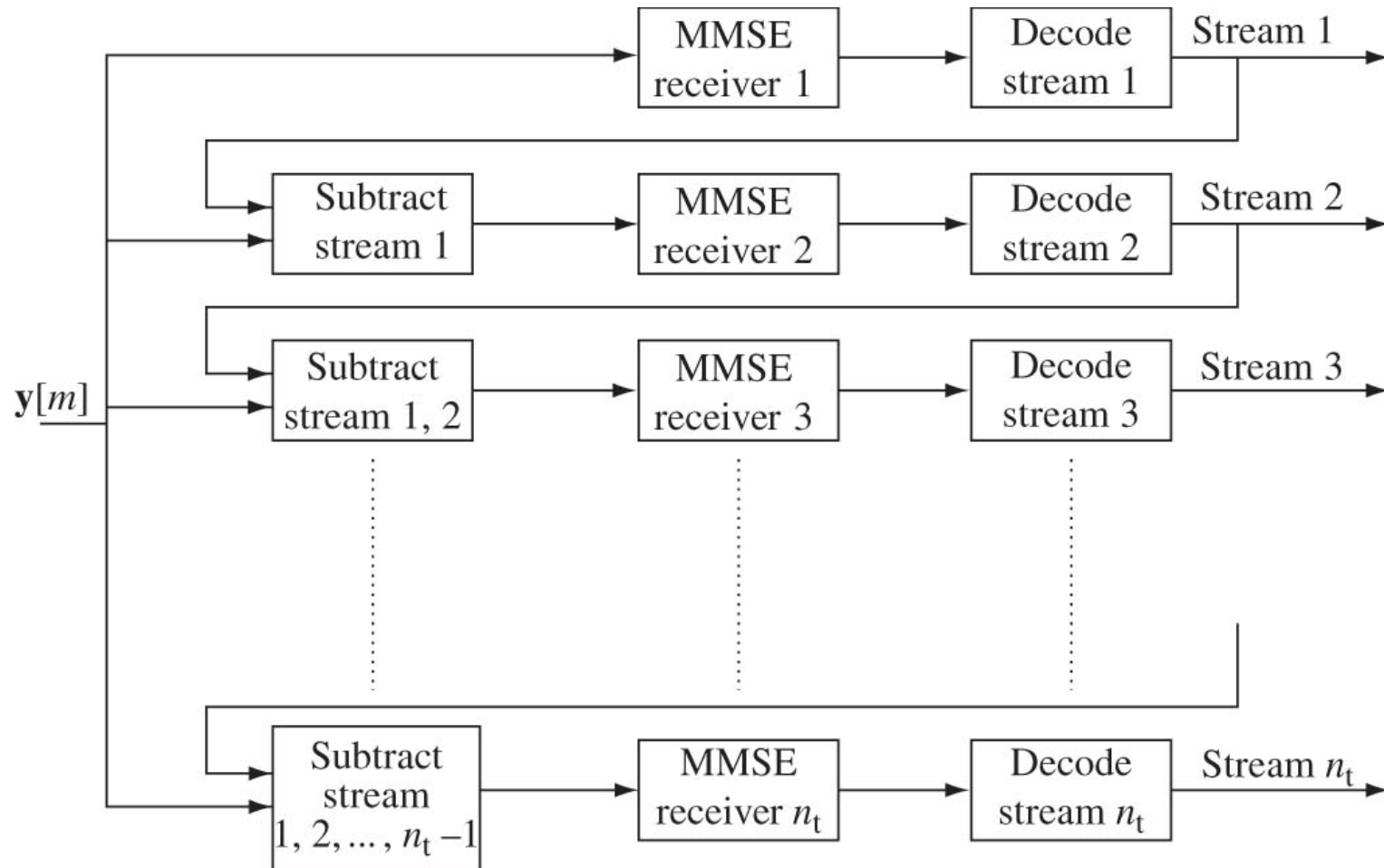


Gap at High SNR

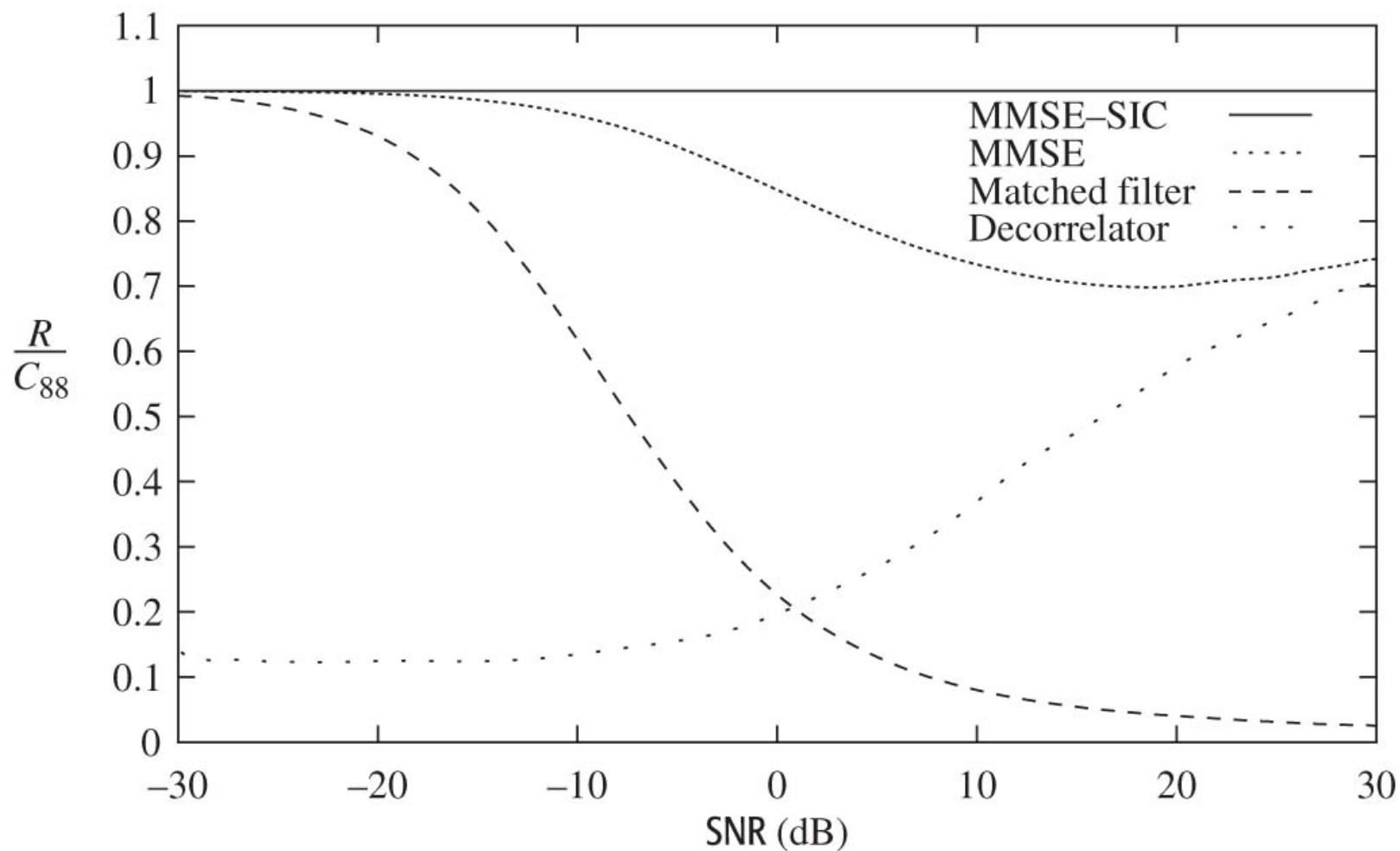
- MMSE improves the performance of decorrelator at moderate and low SNR.
- Does not remove the gap in performance at high SNR
- To remove that gap we have to go to non-linear receivers.



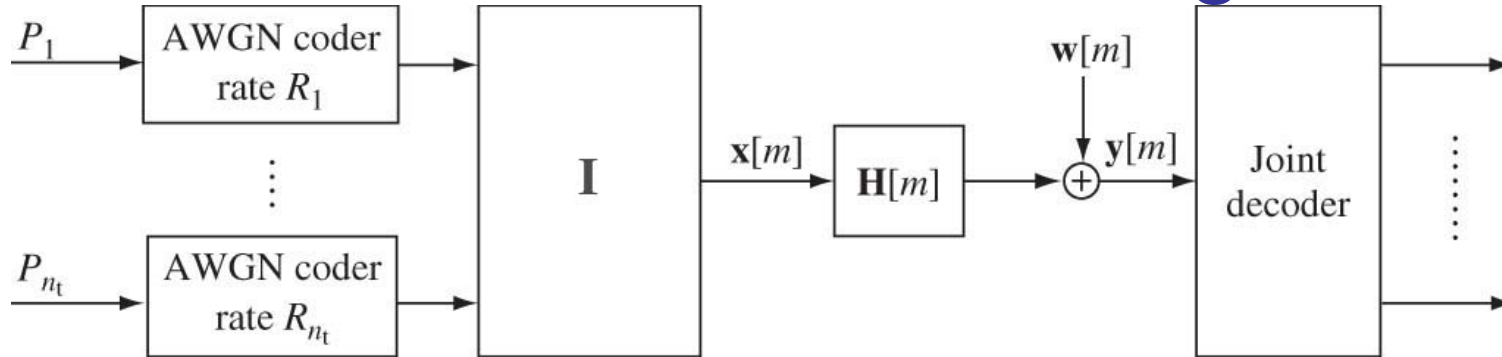
Successive Interference Cancellation



MMSE-SIC Achieves MIMO Capacity

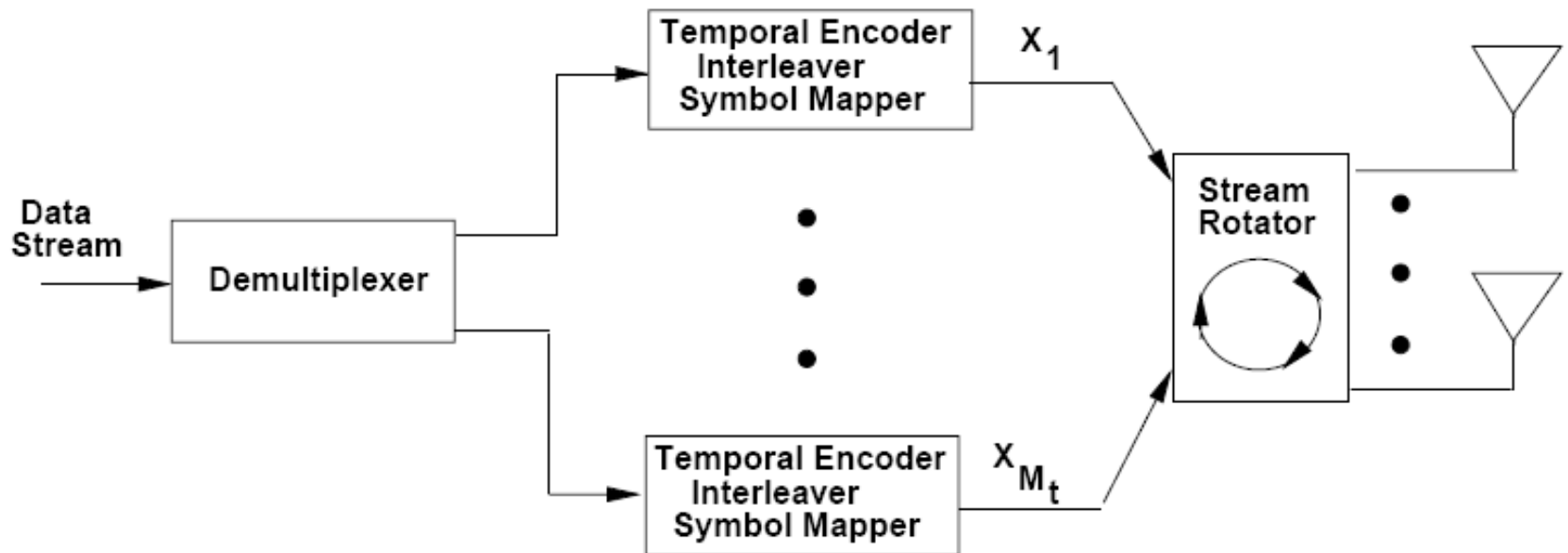


Fast vs Slow Fading

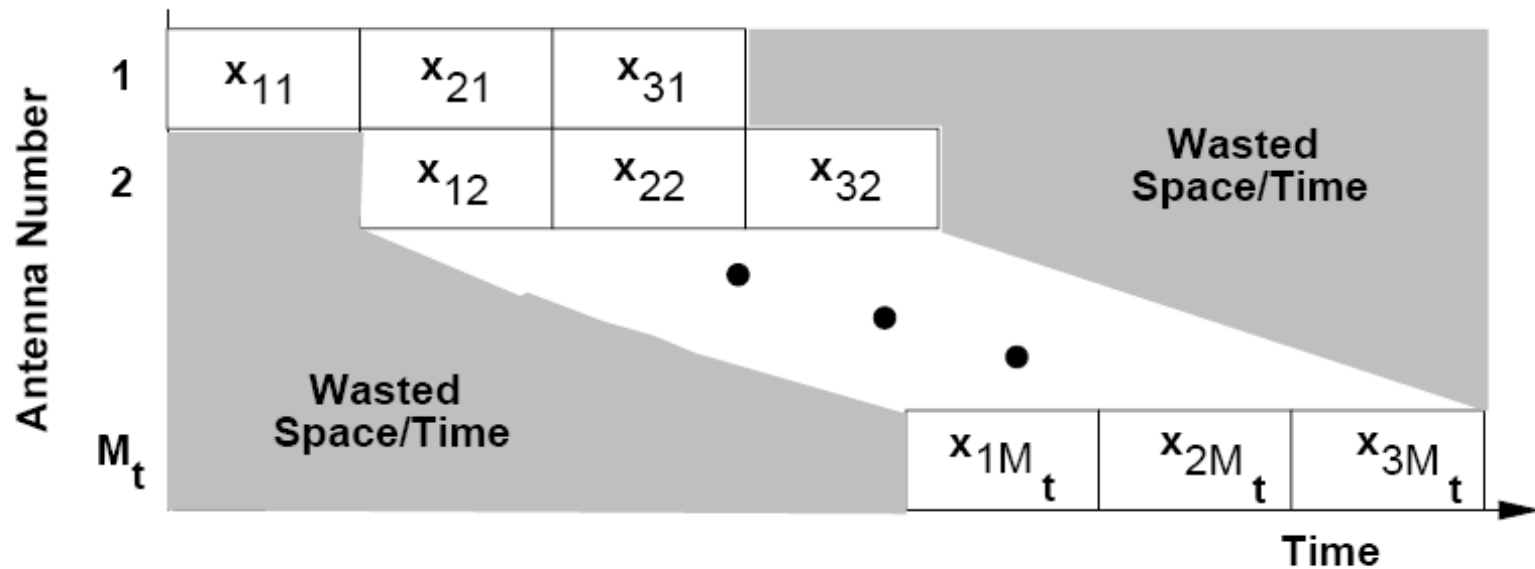


- So far we have focused on the fast fading scenario.
- Can V- BLAST achieve the outage capacity of the slow fading channel?
- No, cannot achieve transmit diversity.
- In fast fading channels, transmit diversity is not important since there is already plenty of time diversity.
- In slow fading channels, there is no time diversity so coding across transmit antennas becomes important-DBLAST.
- Challenge is to combine this with SIC.

D-BLAST

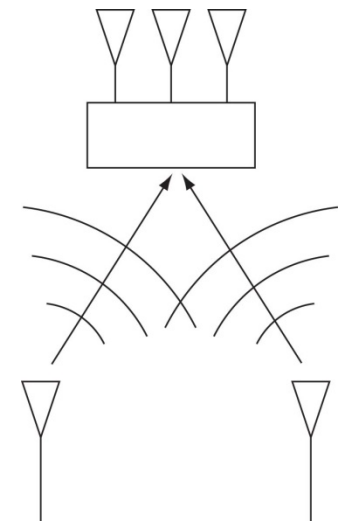


Stream Rotation

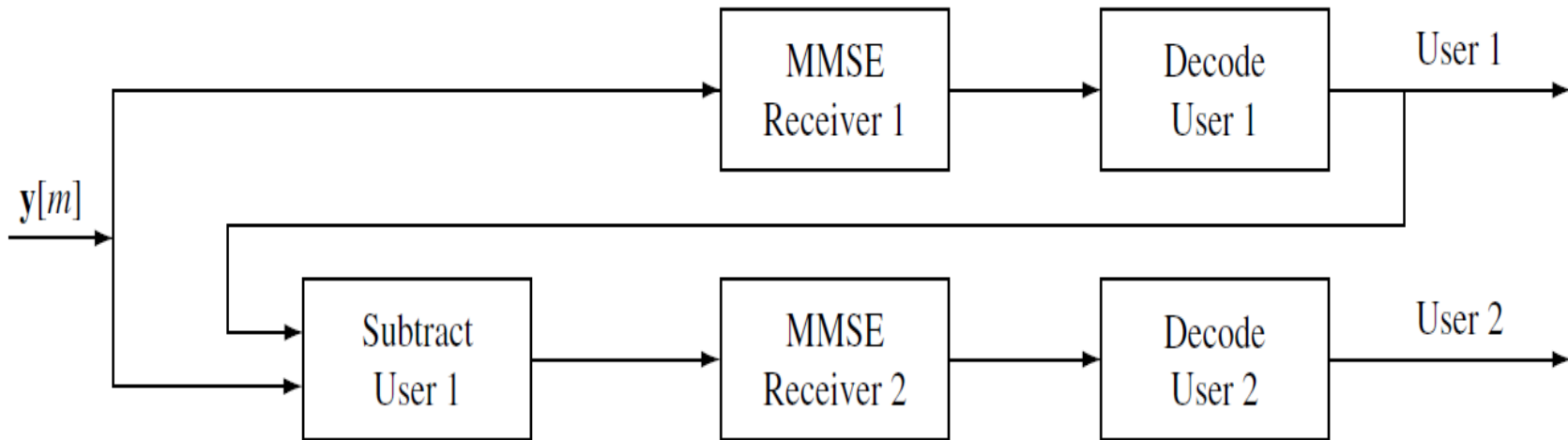


Uplink Architectures

- So far we have considered point-to-point communication.
- But since we are sending independent streams from each transmit antenna, we can use the receiver structures for the uplink with multiple users.
- This is called **space-division multiple access (SDMA)**
- Several simultaneous users can be supported.
- Linear MMSE is also called receive beamforming.

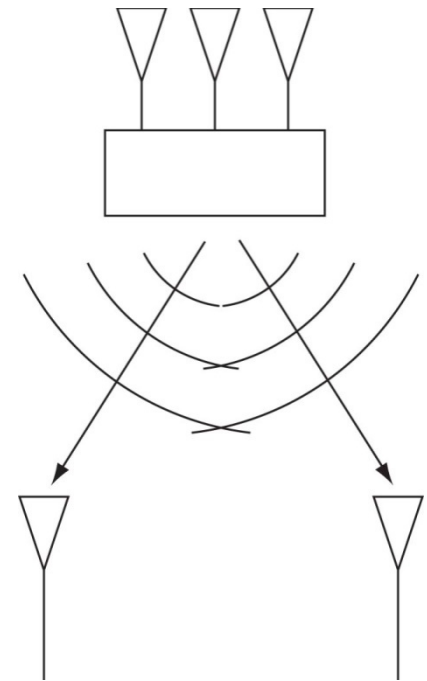


MMSE-SIC Receiver

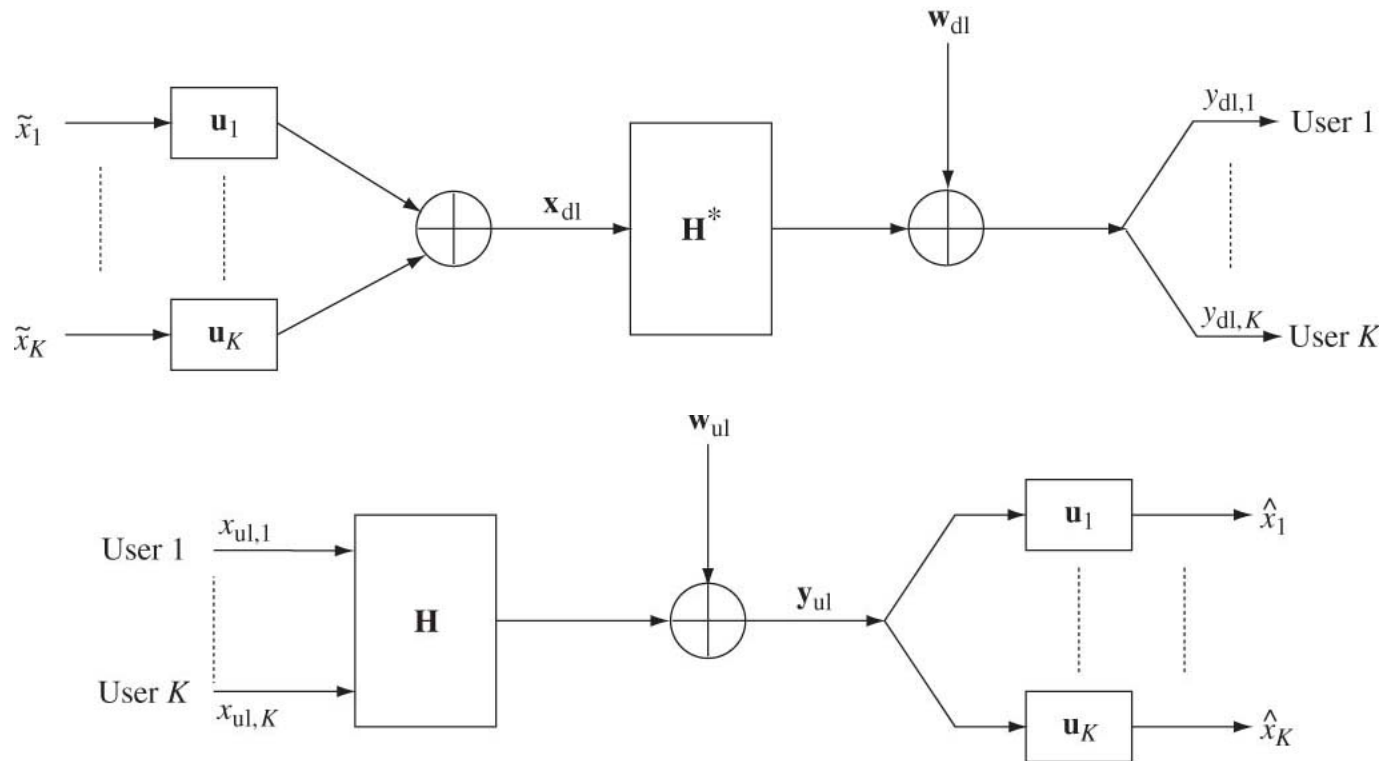


Downlink

- In the uplink, transmitters cannot cooperate, but receiver can jointly process the received signal at all the antennas.
- In the downlink, it is the receivers that cannot cooperate.
- If the transmitter does not track the channel, cannot do SDMA on the downlink.
- If it does, can use techniques reciprocal to the uplink.



Uplink-Downlink Reciprocity



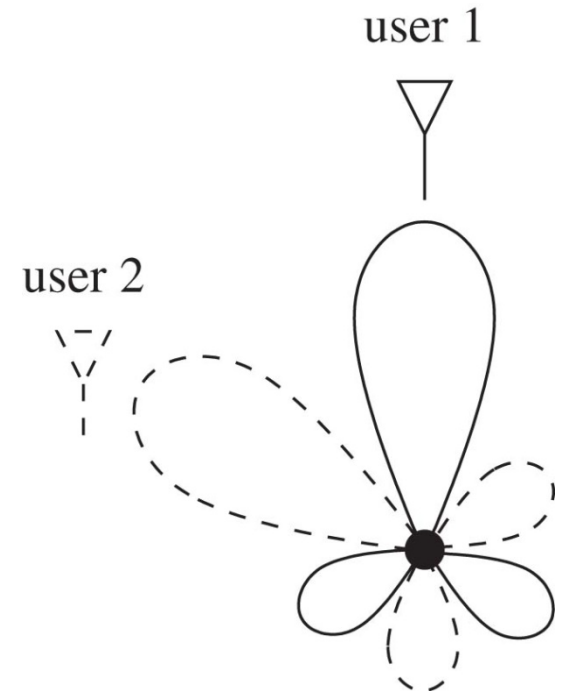
The total power to achieve given SINR requirements is the same in the two links.

Can use MMSE filters in the “virtual” uplink for downlink transmit beamforming.

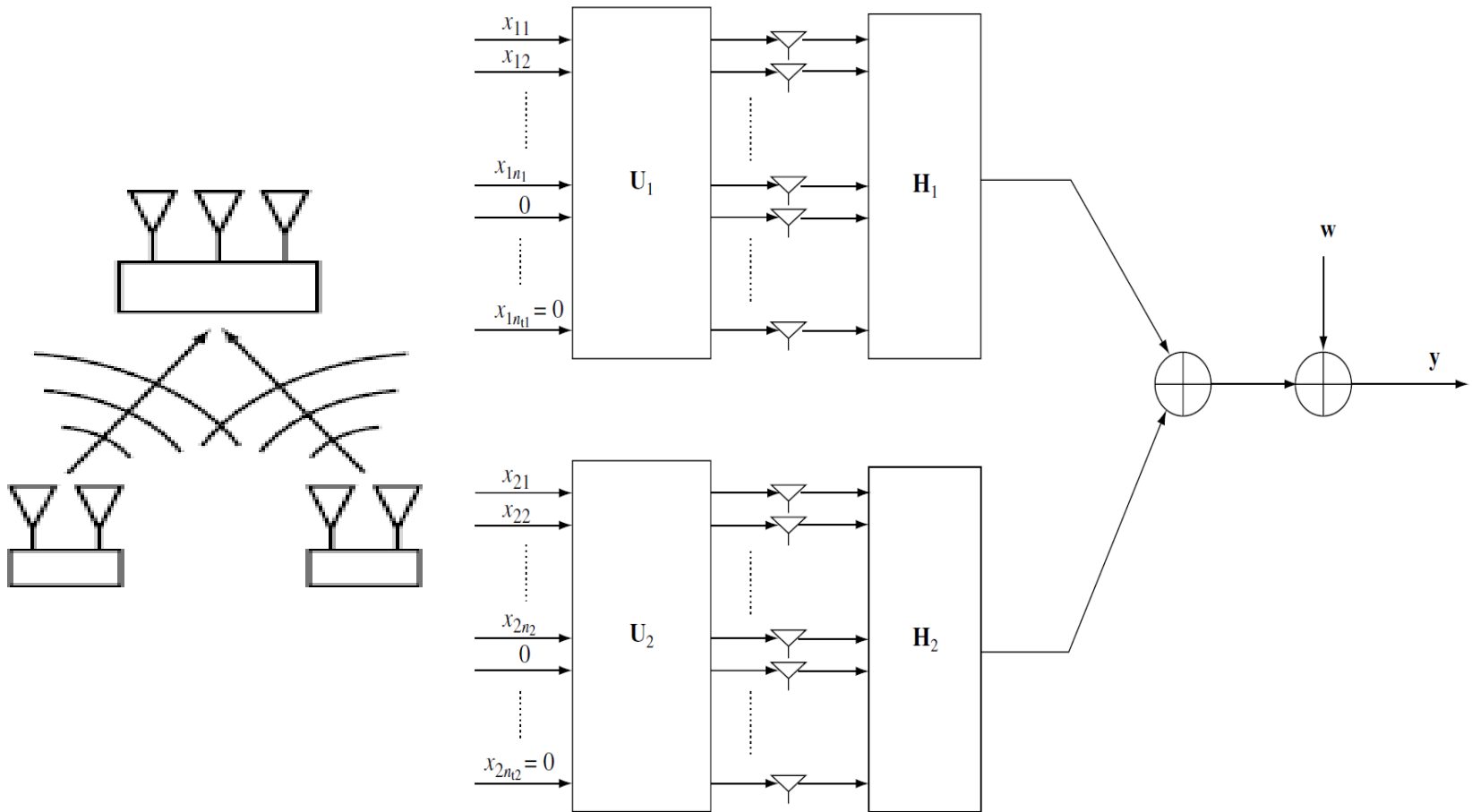
Downlink Transmit Beamforming/Precoding

Can use transmit filter for user 1 that nulls out interference to other users. (downlink decorrelator.)

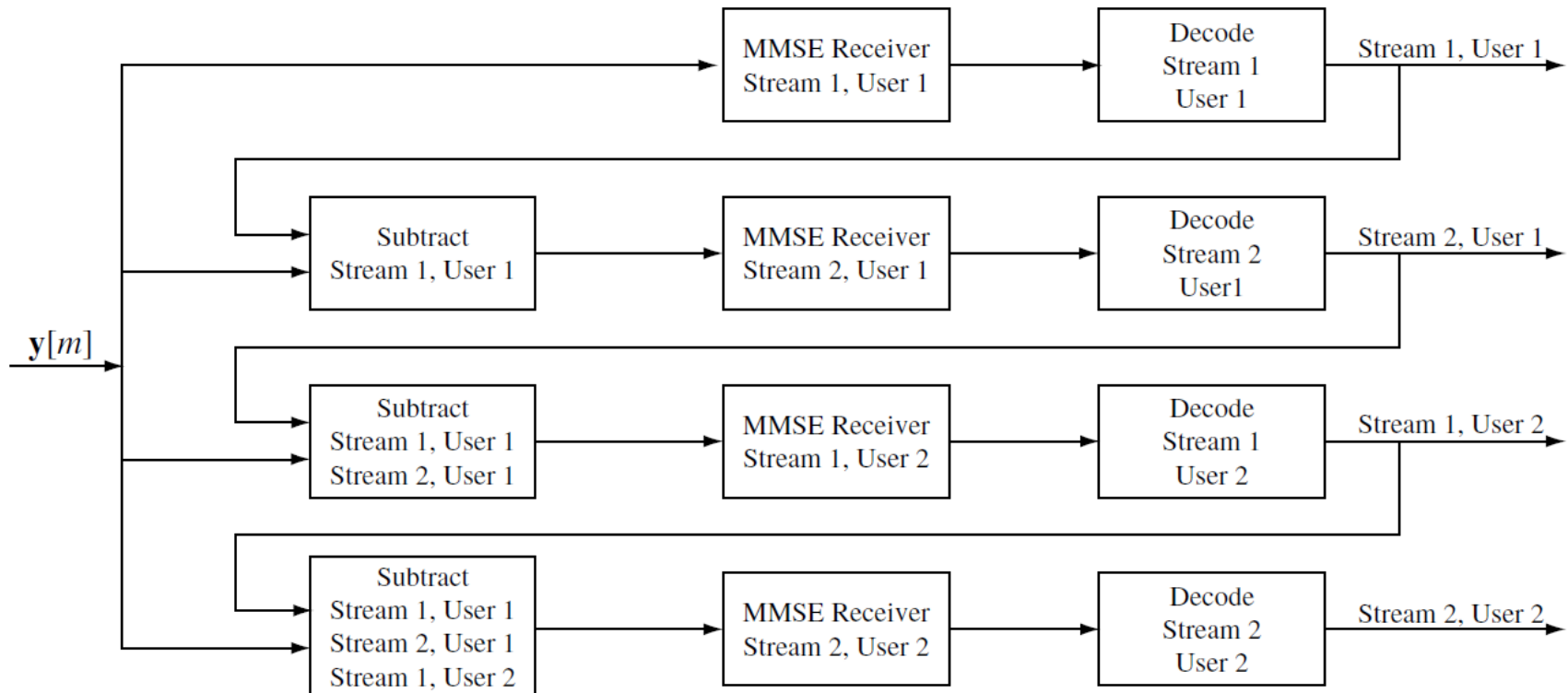
More generally, can optimally balance the energy transferred to the users and the inter-user interference (downlink MMSE)



MIMO-Uplink



Receiver Architecture for MIMO-Uplink



MIMO-Downlink

